

Estimation Multivariate Normal Algorithm with Threshold Convergence

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Abstract:

Keywords: EMNA; EDA; Genetic Algorithms.

Genetic Algorithm (GA) [1]

A **genetic algorithm** is a meta heuristic inspired by the process of natural selection. Genetic algorithms are used to generate (usually) non-optimal but high-quality solutions to optimization and search problems via biologically inspired operators such as *selection*, *crossover*, and *mutation*.

Estimation Distribution Algorithm (EDA)

An **estimation distribution algorithm** is a meta heuristic derived from a GA, where the operations of crossover and crossover are replaced by a probability distribution function (PDF), which is determined by the fittest individuals in the population. This model is used to replace a portion (0% to 100%) of the population. And thus, and EDA converts an optimization problem into finding a fitting PDF for our objective function, in which we expect the PDF to have a higher density near or at local minima.

Estimation Multivariate Normal Algorithm (EMNA)

An **estimation multivariate normal algorithm** is an EDA in which the proposed distribution used to generate the novel populations is a multivariate normal (MVN). We use the fittest individuals mean, variance and covariance to determine the parameters of the MVN. The size of the sample used is given by the *selection coefficient* (sc), which is usually set to 0.30.

Algorithm 1: EMNA

```
1  $X :=$  Initial Population
2 while evals < maxEvals
3    $X_s :=$  Selection( $X$ ,  $sc$ )
4    $[\mu, \Sigma] :=$  ParameterEstimation( $X_s$ )
5    $X :=$  SampleMVN( $\mu$ ,  $\Sigma$ )
6 end
7 return bestSolution
```

EMNA with Threshold Convergence [2]

An **EMNA with Threshold Convergence** proposes using a minimum step size at each generation for the newer solutions. Since there is no parent-child relationship in an EMNA, the step is applied on the parameter space. Specifically, we restrict the norm of the covariance matrix generated from the elite individuals. We define the *umbral* as

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$$\text{umbral} = \alpha \times \text{diagonal} \times \left(\frac{\text{maxEvals} - \text{evals}}{\text{maxEvals}} \right)^\gamma.$$

We use this number to perform the following action on the covariance matrix (Σ):

$$\Sigma := \text{umbral} * \frac{\Sigma}{\|\Sigma\|}.$$

This prevents the EMNA from prematurely converging.

Results

We used 3 test functions [3], [4], [5] on the algorithm, for $d = 2, 5$:

$$\text{Sphere: } f(x) = \sum_{i=1}^d x_i^2$$

$$\text{Griewank: } f(x) = \sum_{i=1}^d \frac{x_i^2}{4000} - \prod_{i=1}^d \cos\left(\frac{x_i}{\sqrt{i}}\right) + 1$$

$$\text{Ackley: } f(x) = -a \exp\left(-b \sqrt{\frac{1}{d} \sum_{i=1}^d x_i^2}\right) - \exp\left(\frac{1}{d} \sum_{i=1}^d \cos(cx_i)\right) + a + \exp(1)$$

We used a population of $n = 1000$, with a maximum of $100000 \times d$ functions evaluations (one more zero than in the original paper).

Function	Best	Worst	Mean	Median	StdDev
Ackley	9.97×10^{-3}	1.16×10^{-1}	4.63×10^{-2}	3.92×10^{-2}	2.96×10^{-2}
Griewank	3.44×10^{-3}	1.50×10^{-2}	8.50×10^{-3}	8.59×10^{-3}	3.63×10^{-3}
Sphere	7.46×10^{-6}	9.79×10^{-5}	6.39×10^{-5}	7.86×10^{-5}	3.16×10^{-5}

Table 1: Statistics of EMNA for 10 randomly seeded runs on $d = 2$

Function	Best	Worst	Mean	Median	StdDev
Ackley	5.05×10^{-1}	1.79×10^0	1.13×10^0	1.27×10^0	4.35×10^{-1}
Griewank	1.64×10^{-1}	4.38×10^{-1}	2.52×10^{-1}	2.52×10^{-1}	7.24×10^{-2}
Sphere	2.86×10^{-4}	2.12×10^{-3}	1.37×10^{-3}	1.42×10^{-3}	5.03×10^{-4}

Table 2: Statistics of EMNA for 10 randomly seeded runs on $d = 5$

Conclusions

We observed that EMNA struggled to find the true minimum of the Ackley function for both test cases. The other two functions we closer to converging. This makes sense, as the Ackley function is nearly flat outside the origin, with many local minima, while the origin is a steep drop-off to the true global minimum. The Griewank function has the form of a paraboloid in 2D, where locally, the curve looks like a cosine function in 2D. This leads to many local minima, while the true global minimum is in the origin. Compared to Ackley, this functions local minima converge towards the global minimum, although it is possible to get stuck in a local minimum. Finally, the sphere function is quite simple, allowing for a fast convergence to the global minimum.

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